



Pelita Bangsa
Academy

Introduction to Lisk Blockchain & IDRX Stablecoin

BUILDing with IDRX Stablecoin on Lisk Bootcamp

Hi, I'm **Heinrich Wisesa** from Indonesia 

5+ years

Experience in the IT industry



**Alumni Blockchain Developer
Bootcamp Cohort 2**



**Alumni BUIDL Web3 Ideas With
Lisk Cohort 1**



**Institut Sains dan Teknologi
Terpadu Surabaya**

S.Kom. | Information Technology | GPA 3.99

Working Experience



PT Bank Central Asia

Sr IT Specialist | IT Team Lead
for myBCA Bisnis

Teaching Experience



Pelita Bangsa Academy

**Mall of Indonesia, Lantai 2, Unit 2F-A7B,
Jalan Boulevard Barat Raya No. 1,
Kelapa Gading Jakarta Utara 14240.**



”

Upskill dan Reskill Talenta Teknologi Indonesia di bidang Blockchain sehingga kalian bisa bersaing secara global.



SESI

01

**Introduction to Lisk Blockchain
& IDRX Stablecoin**

HARI INI

02

Smart Contract Development on Lisk

26 Mar

03

Interacting with IDRX Stablecoin

9 Apr

SESI

04 Building Frontend UI for Web3 DApp

**Extra
Class**

Introduction to Xellar Kit

05 Integrating Xellar Kit with Smart Contract

06 Brainstorming Web3 Project Ideas

16 Apr

17 Apr

23 Apr

30 Apr

Agenda

**Goal of this
Bootcamp**

Materi

Hands On

Q&A

Wrap Up

Foto Bersama

Recommendation



Physical Notebook



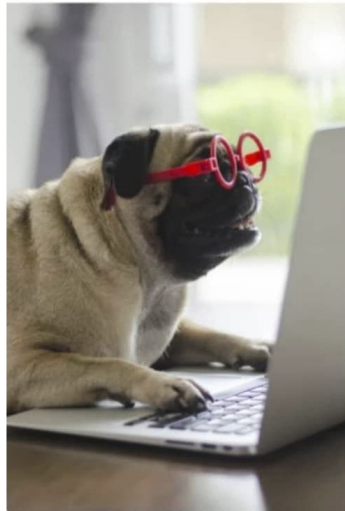
<https://notion.com>

Be Active

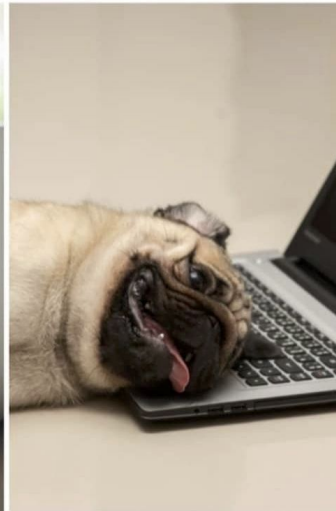


Open Your Camera

Zoom meeting,
with video

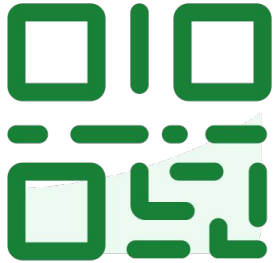


Zoom meeting,
audio only





slido



Join at slido.com
#235689



Click **Present with Slido** or install our
instructions for participants while presenting.

to display joining



Apakah kamu punya background IT?

① Click **Present with Slido** or install our poll while presenting.

to activate this



Seberapa mahir kamu dalam teknologi Blockchain?

① Click **Present with Slido** or install our poll while presenting.

to activate this

A

Goal of this Bootcamp



Pelita Bangsa Academy Lisk 1000 IGNITE
BlockDevId X IDR X Xellar

* ♦ *

BUIDLing with IDRX Stablecoin on Lisk Bootcamp!

* ♦ *

 GRATIS!

LIVE via Zoom

→ Swipe untuk daftar

The goal of this bootcamp:

- Learn about Lisk Blockchain Technology
- Come up with Web3 Application Ideas
- #BuildOnLisk Blockchain
- Participate Lisk Builders Challenge Round 1
- Build successful Web3 Startup !!



Fund Tracking Management





Lisk

BlockDevId



IDRX



Xellar



LISK

Builders
Round 1
Challenge





Project Requirement:

1. Smart Contract deployed on **Lisk Sepolia Testnet** or **Lisk Mainnet** and verified on **Blockscout Explorer**
2. Integrate your dApp with **IDRX Stablecoin** on **Lisk Mainnet** or **Lisk Sepolia Testnet**
3. Build your dApps with **Xellar Kit**
4. Live Project URL

IDRX Address on Lisk Mainnet: 0x18Bc5bcC660cf2B9cE3cd51a404aFe1a0cBD3C22

IDRX Address on Lisk Sepolia Testnet: 0xD63029C1a3dA68b51c67c6D1DeC3DEe50D681661

Prize:

3,000 USD (1st Winner)

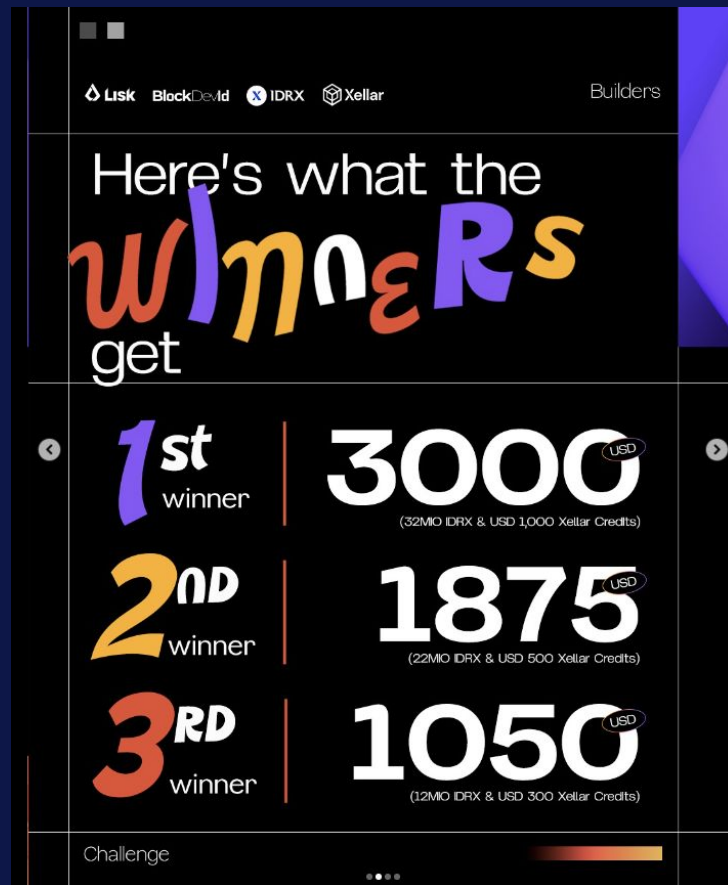
(32 Million IDR^X & USD 1,000 Xellar Credits)

1,875 USD (2nd Winner)

(22 Million IDR^X & USD 500 Xellar Credits)

1,050 (3rd Winner)

(12 Million IDR^X & USD 300 Xellar Credits)

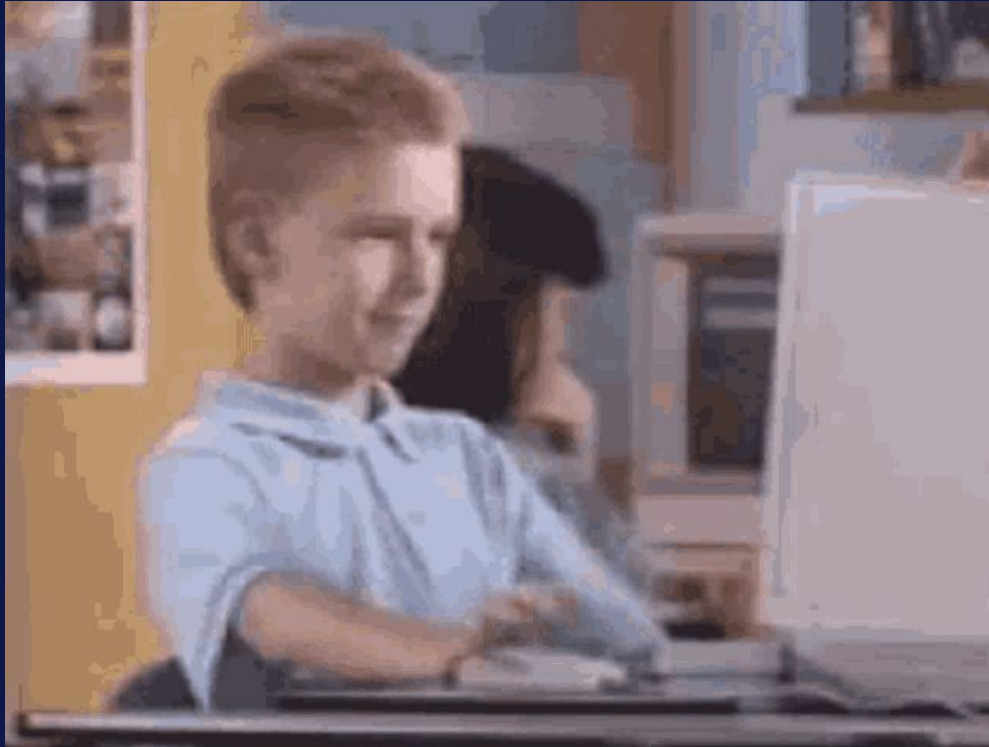


Here's what the winners get

1 st winner	3000 ^{USD} (32MIO IDR ^X & USD 1,000 Xellar Credits)
2 nd winner	1875 ^{USD} (22MIO IDR ^X & USD 500 Xellar Credits)
3 rd winner	1050 ^{USD} (12MIO IDR ^X & USD 300 Xellar Credits)

Challenge

Not bad



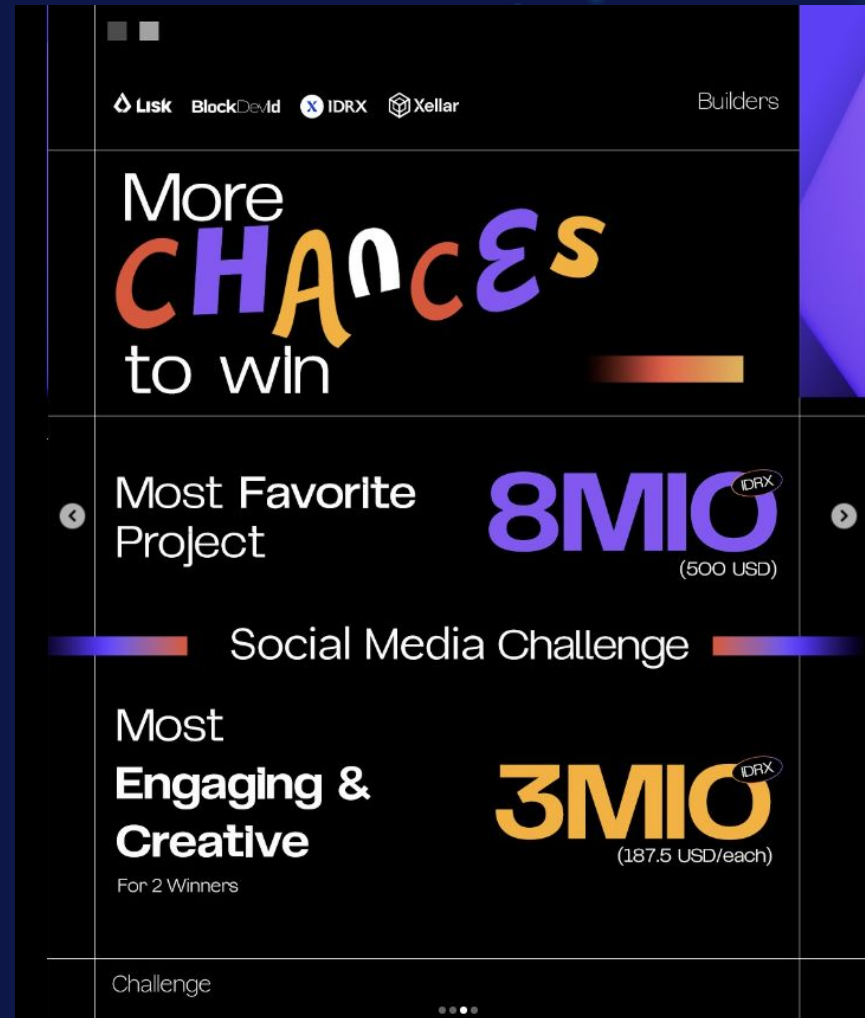


Prize:

8 Million IDR^X (Most Favorite Project)

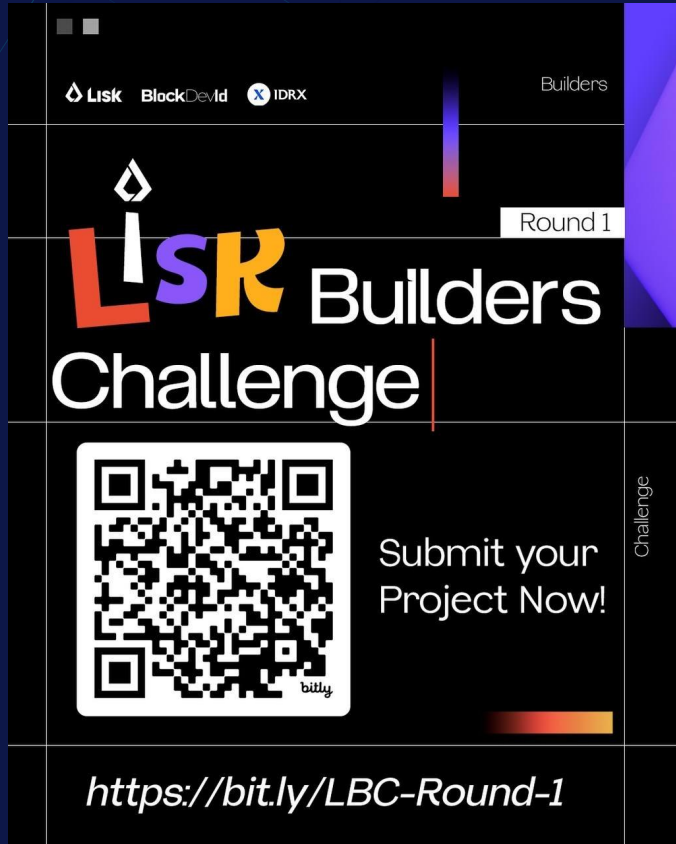
3 Million IDR^X (SocMed Challenge Winner #1)

3 Million IDR^X (SocMed Challenge Winner #2)



YESSSS!!





The poster for the Lisk Builders Challenge Round 1 features a dark background with a purple and blue gradient on the right side. At the top left, there are logos for Lisk, BlockDevId, and IDRX. The word "Builders" is in the top right. The main title "Lisk Builders Challenge" is prominently displayed in the center, with "Lisk" in large, colorful letters (red, purple, orange) and "Builders Challenge" in white. Below the title is a large QR code with the text "Submit your Project Now!" to its right. At the bottom, the URL <https://bit.ly/LBC-Round-1> is shown. A vertical label "Challenge" is on the right edge of the poster.

Timeline:

- Hackathon Start: March 8, 2025
- Kickoff AMA: March 12, 2025
- Workshop: March 19 – April 30, 2025
- Submission Deadline: May 17, 2025
- Announcement of Finalists: May 24, 2025
- Finalists Coaching: May 26–30, 2025
- Demo Day (Jakarta – Hybrid): May 31, 2025



B

Pelita Bangsa Academy Resources



Pelita Bangsa Academy Lisk 1000 IGNITE
BlackDevid X IDRX Xellar

* ♦ *

BUIDLing with IDRX Stablecoin on Lisk Bootcamp!

* ♦ *

 GRATIS!

LIVE via Zoom

→ Swipe untuk daftar

Pelita Bangsa Academy Resources:

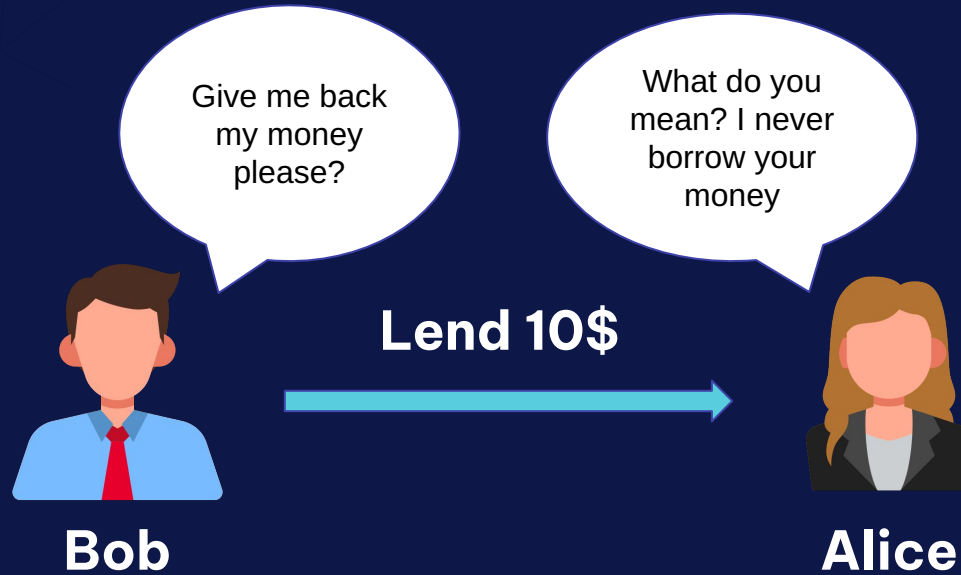
- **LMS (Learning Management System):**
<https://lms.pelitabangsa.co.id/>
(Class Recording, PDF Material & Other Links)
- **Discord Link:**
<https://discord.gg/Jquvw5DqEv>
(Discussion on #lisk channel)

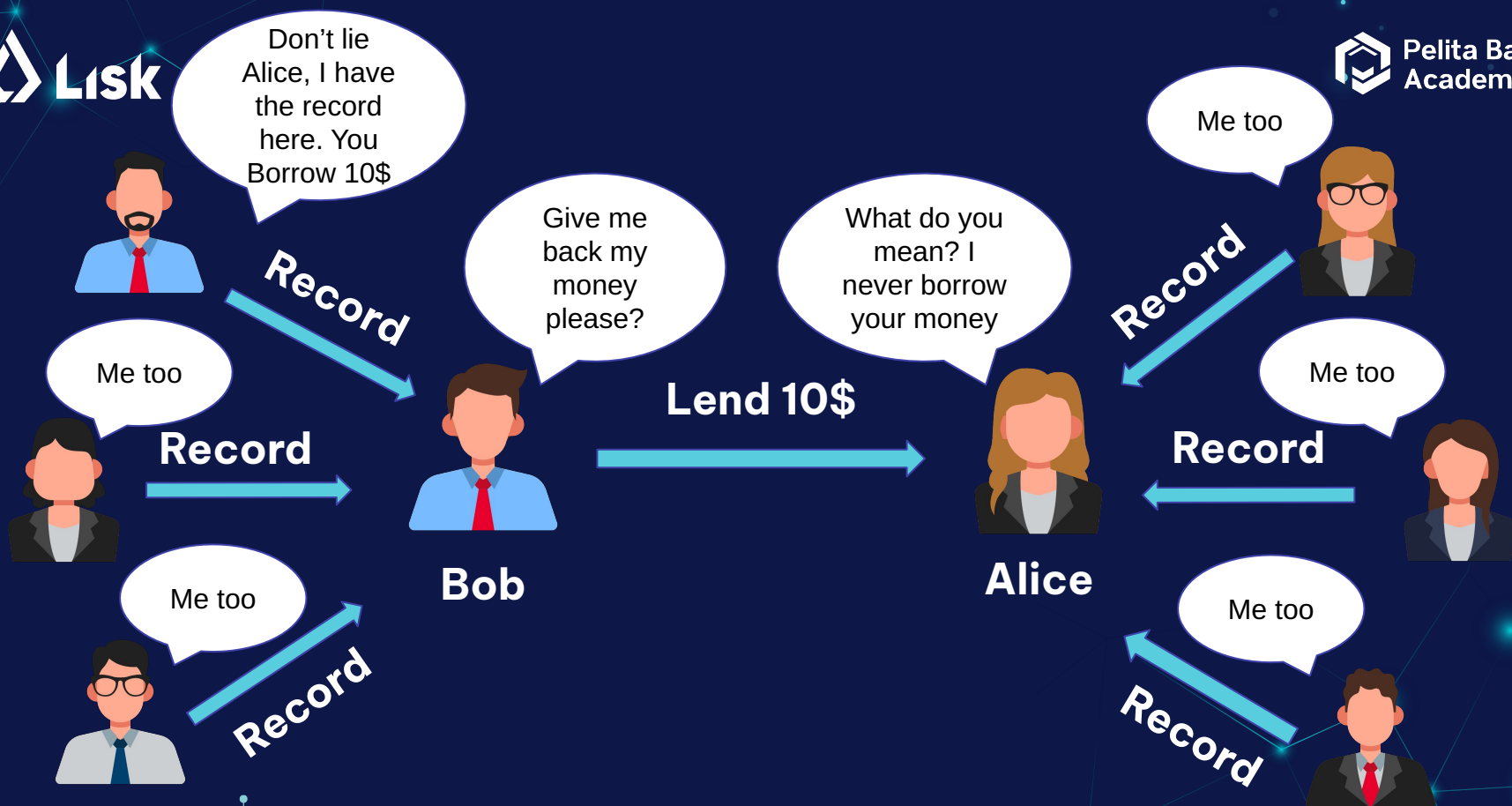
01

What is Blockchain?

What is Blockchain?

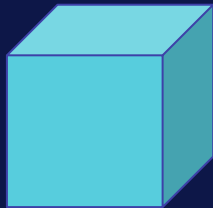
Blockchain is a decentralized database that records transactions across a network of computers (nodes).





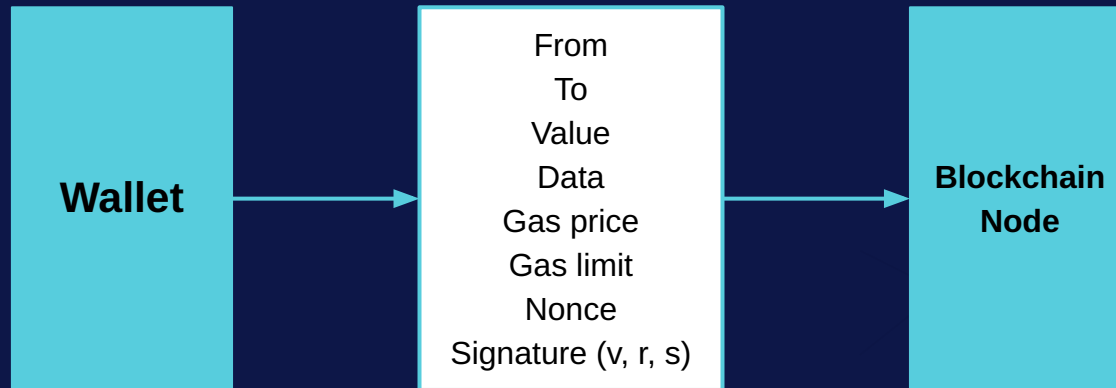
What is a Block?

A block contains **a set of validated transactions**, linked to the previous block in the chain, creating a **secure** and **immutable** record of information within the blockchain network

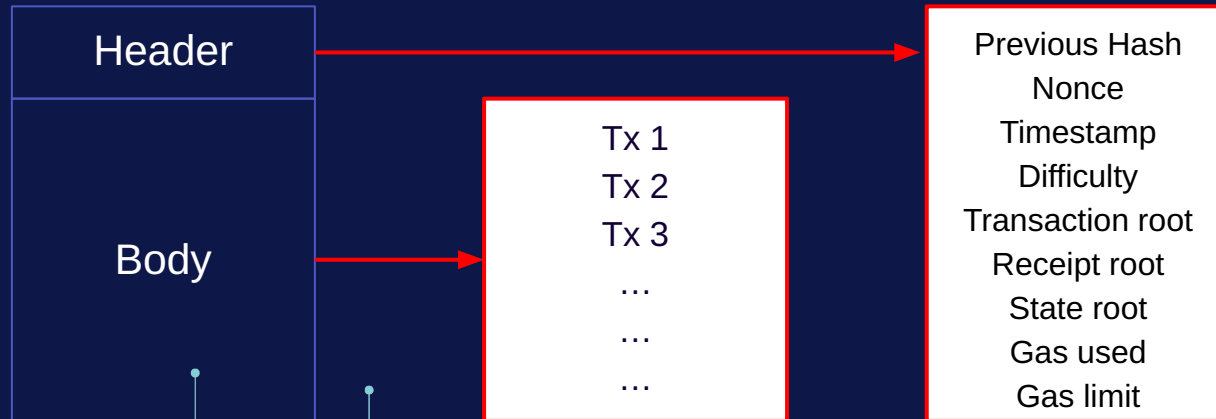


Tx 1: Bob send Alice 1 BTC
Tx 2: Alice send Jim 2 BTC
Tx 3: Jim send Bob 3 BTC
....

Transaction

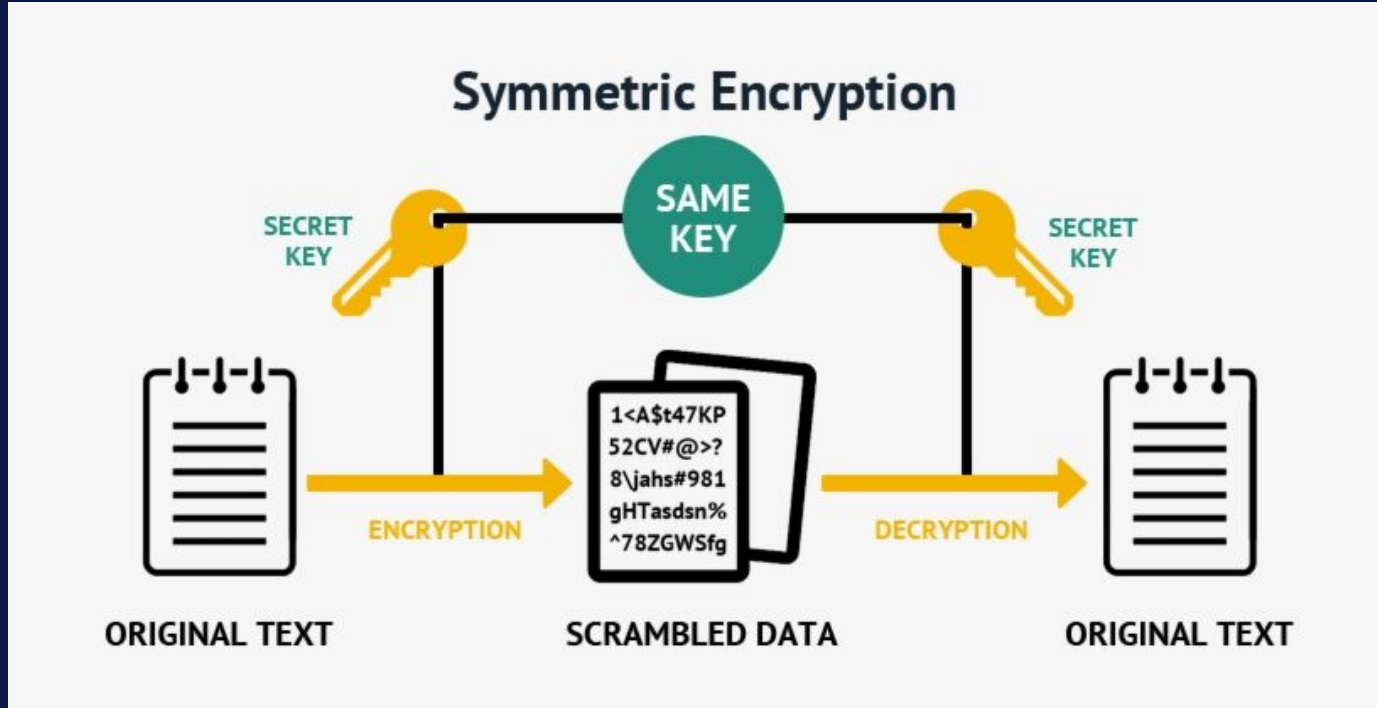


Block

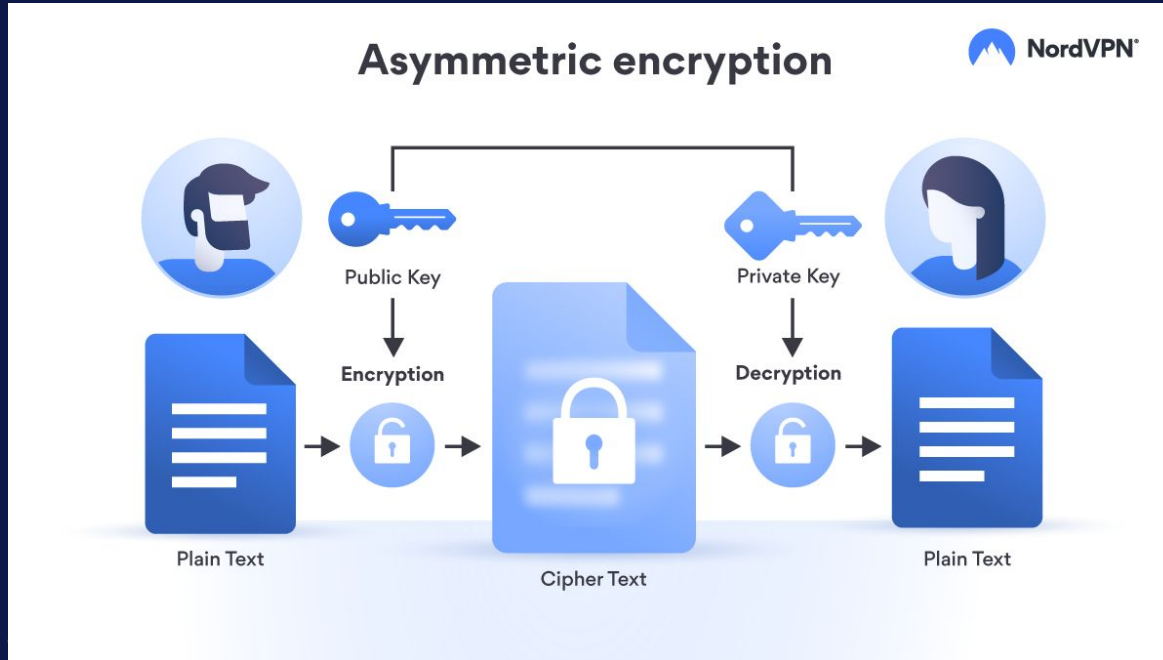


Where do wallet address come from?

Symmetric Key Encryption



Asymmetric Key Encryption



<https://nordvpn.com/blog/what-is-asymmetric-cryptography/>

Hash

Hello world



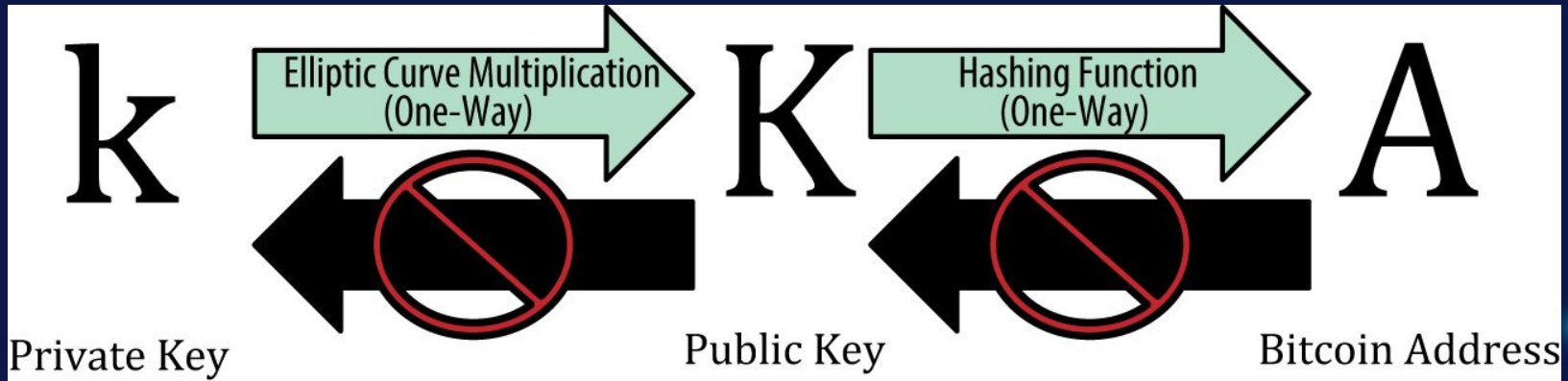
ed6c11b0b5b80896
0df26f5bfc471d04c
1995b0ffd2055925a
d1be28d6baadfd

Pelita Bangsa Academy



056a08627e0be4ba
47630341156d3a8e
291da95a9275fd9b
1485415ed5cb1b06

Asymmetric Cryptographic Keys



Source: Andreas M. Antonopoulos,
<https://github.com/bitcoinbook/bitcoinbook/blob/develop/ch04.asciidoc>

Wallet

4737c03c3eace1e6ada59a
ba789fb0c7bb0d84acb1536
e9141fc0293a615f8a1

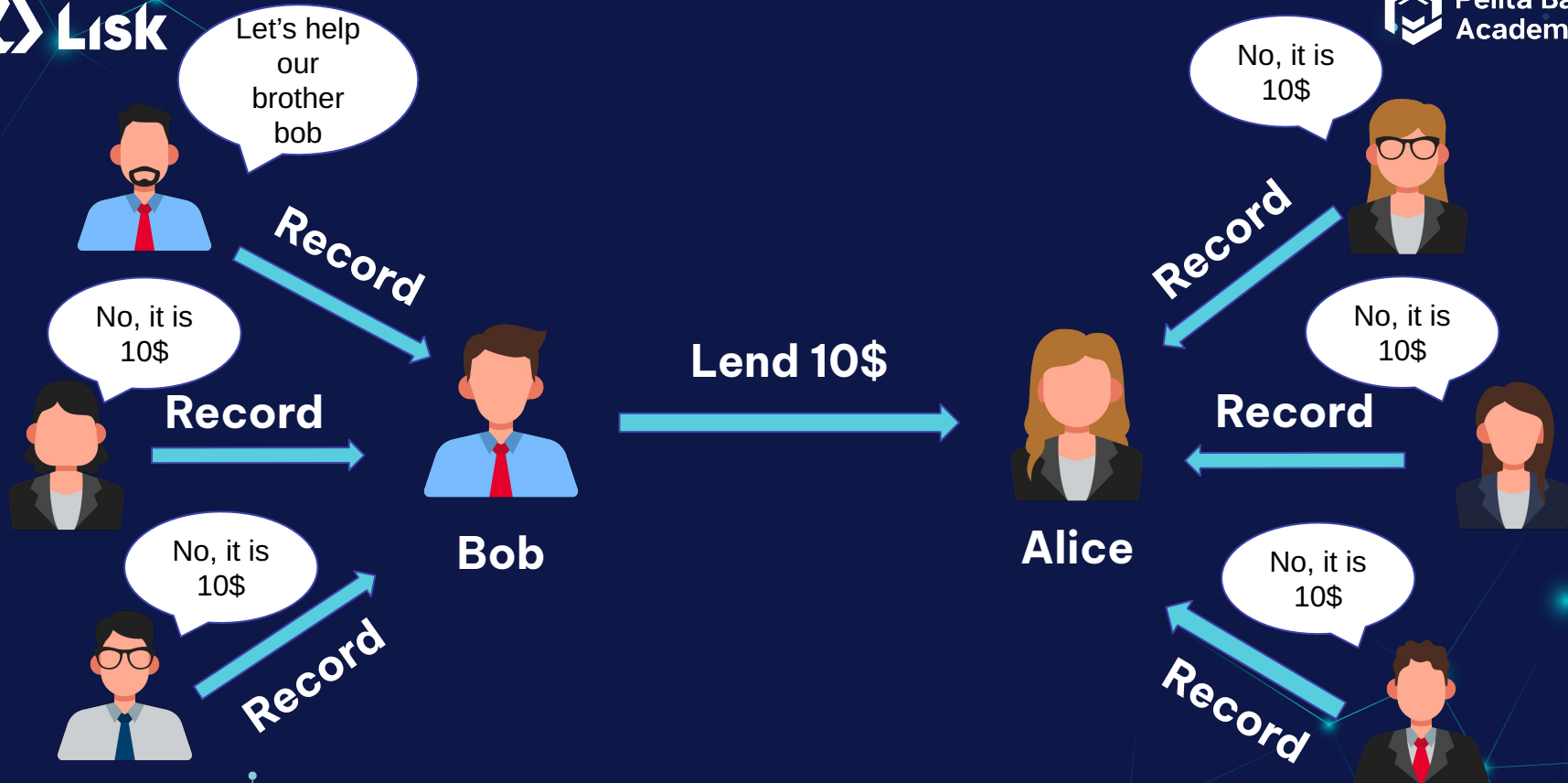
Public key



**0xaaA86fEEA60960bb89e43
f311496911dc585722a**

Address

**What if someone tamper
previous transaction?**



Consensus Mechanism



A set of rules or protocols that allows all nodes on a decentralized network to agree on the validity of transactions and the current state of the blockchain



PoW (Proof of Work)



Miners compete to solve complex mathematical problems to validate transactions and add new blocks to the blockchain



PoS (Proof of Stake)



Validators are chosen to create new blocks and validate transactions

02

Implementasi Teknologi Blockchain

Types of Cryptocurrencies

Coin

- Built into a blockchain - **native to its own blockchain**
- Either **mined** (in PoW) or **earned** (in PoS)



Token

- Built on top of an existing blockchain - **not native**
- Relatively **easy** to create
- Usually **Distributed** by project owner



Cryptocurrency



Cryptos: 2.4M+ Exchanges: 770 Market Cap: \$2.57T ▼ 0.75% 24h Vol: \$87.56B

Implementasi: Stablecoin | Utility | Governance | Security | Equity Tokens

Sumber: coinmarketcap.com

- 1  Bitcoin BTC
- 2  Ethereum ETH
- 3  Tether USDT
- 4  BNB BNB
- 5  Solana SOL

NFT (Non Fungible Token)



Bored Ape Yacht Club

Market Cap: USD 712.19M



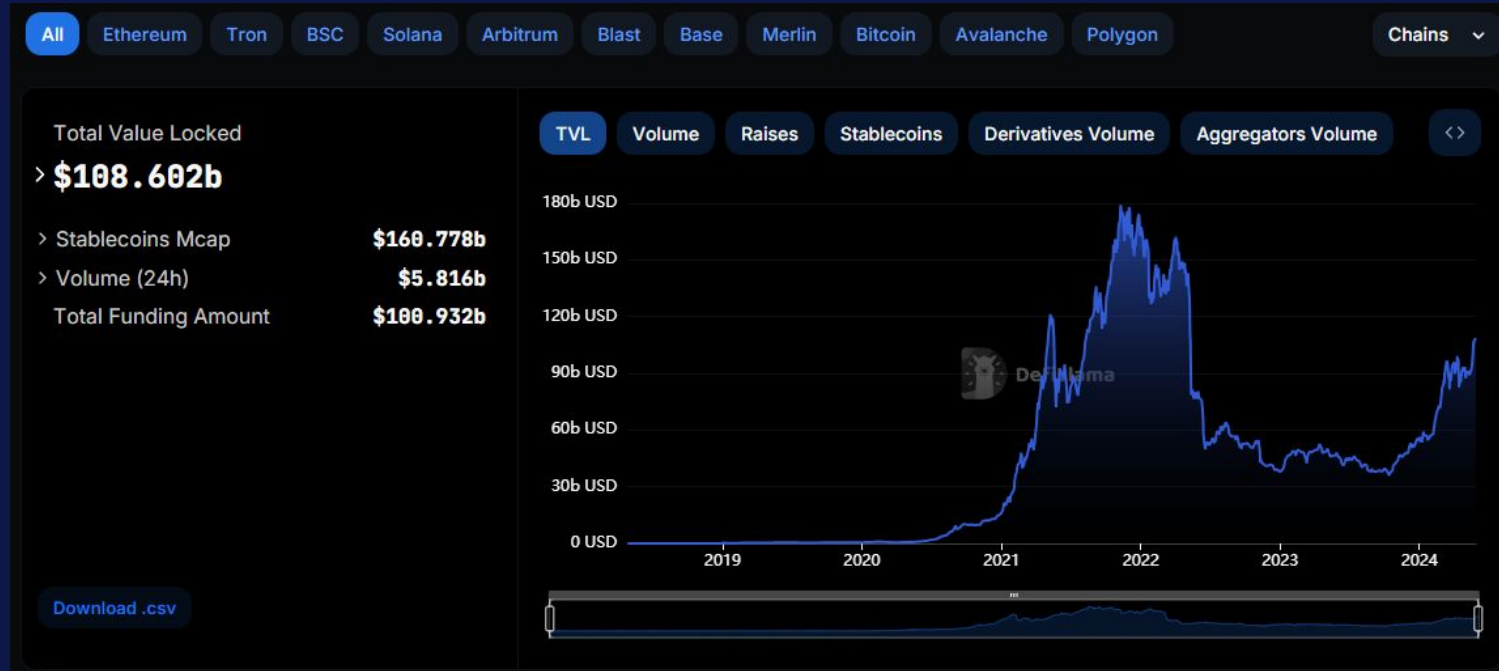
CryptoPunks

Market Cap: USD 2.42B

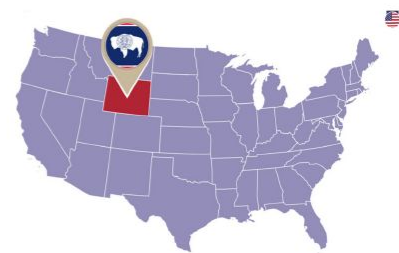
Sumber: nftvaluations.com

DeFi (Decentralized Finance)

Implementasi: Lending and Borrowing | Decentralized Exchanges (DEXs) | Staking | Yield Farming



Sumber: defillama.com



DAOs

(Decentralized Autonomous Organization)

Policy

State Lawmaker Explains Wyoming's Newly Passed DAO LLC Law

Wyoming DAO LLCs have to be domiciled in the state, which could be a point of confusion for denizens of the decentralized web.

By Nate DiCamillo ⌚ Apr 23, 2021 at 1:21 a.m. Updated Sep 14, 2021 at 7:45 p.m.

Policy

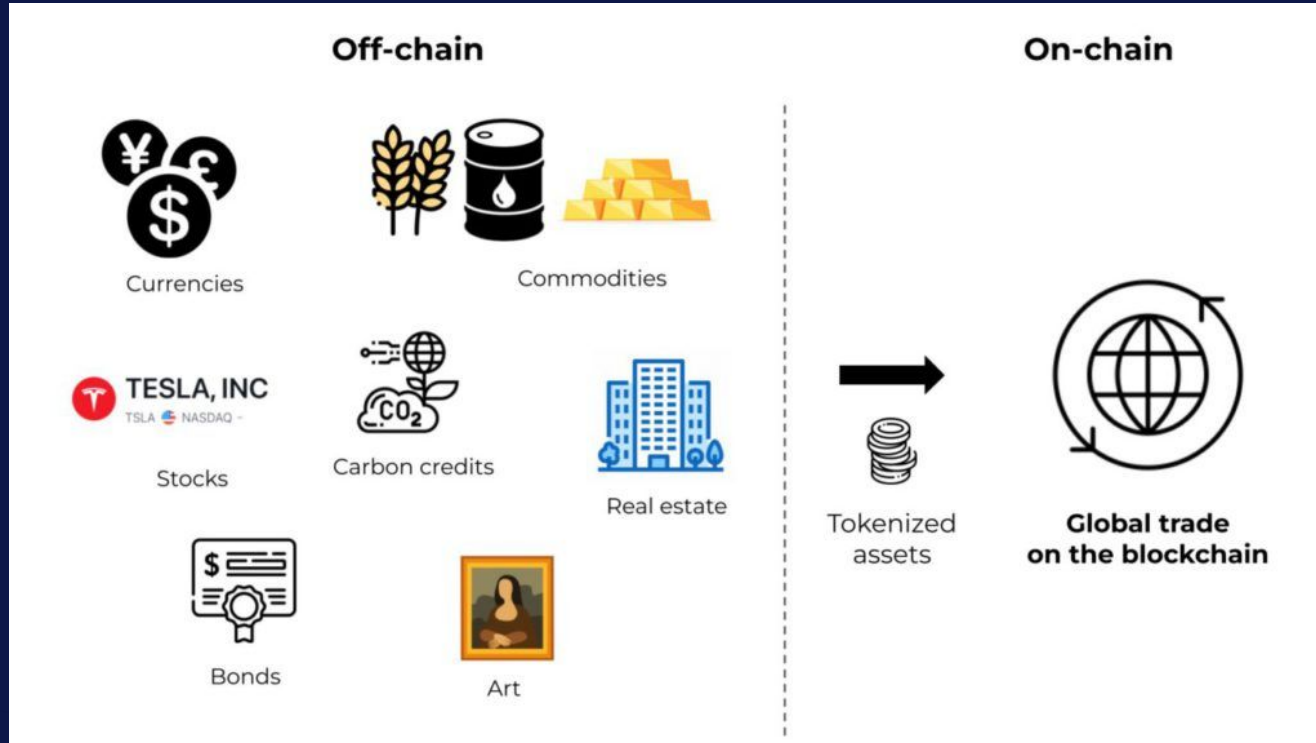
Wyoming Grants DAOs New Legal Structure

The governor signed a law setting up a legal status – "decentralized unincorporated nonprofit associations" – for DAOs there, and one investment firm called Wyoming an "oasis."

By Jesse Hamilton ⌚ Mar 9, 2024 at 5:06 a.m. Updated Mar 9, 2024 at 5:08 a.m.



RWA

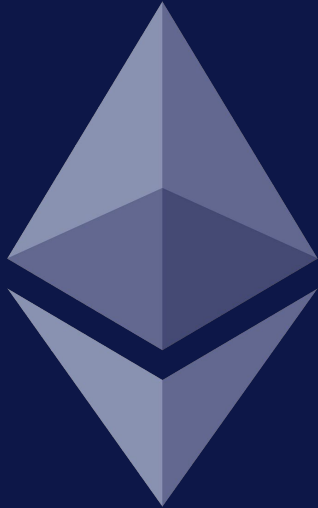


Sumber: Apollo Crypto

03

Ethereum & EVM

Ethereum in a Nutshell



- Created in 2015 by Vitalik Buterin
- Supports Turing-complete smart contracts
- Entirely open-source
- Proof-of-Stake since Ethereum 2.0
- Account/Balance model
- Block reward adjustments, but no halving mechanism

Smart Contract



```
1 // SPDX-License-Identifier: UNLICENSED
2 pragma solidity ^0.8.0;
3
4 contract Counter {
5     uint256 private _count;
6
7     function getCount() public view returns (uint256) { 2415 gas
8         return _count;
9     }
10
11     function increment() public { infinite gas
12         _count += 1;
13     }
14 }
```

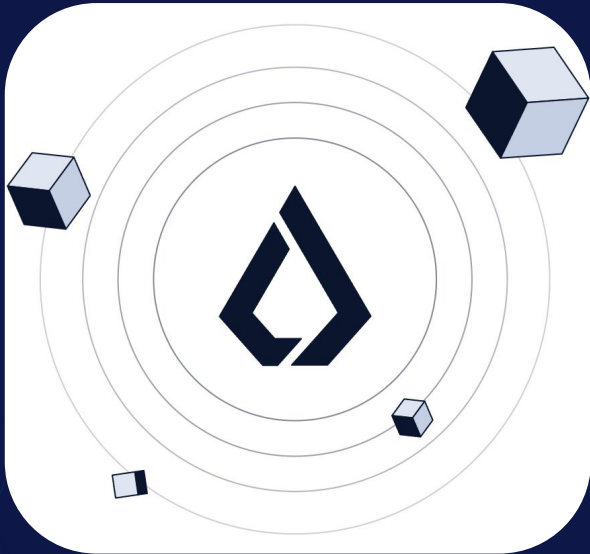
EVM (Ethereum Virtual Machine)

- A component of the Ethereum blockchain that enables the execution of smart contracts
- EVM uses a gas mechanism to measure and limit the computational work required to execute transactions and smart contracts

04

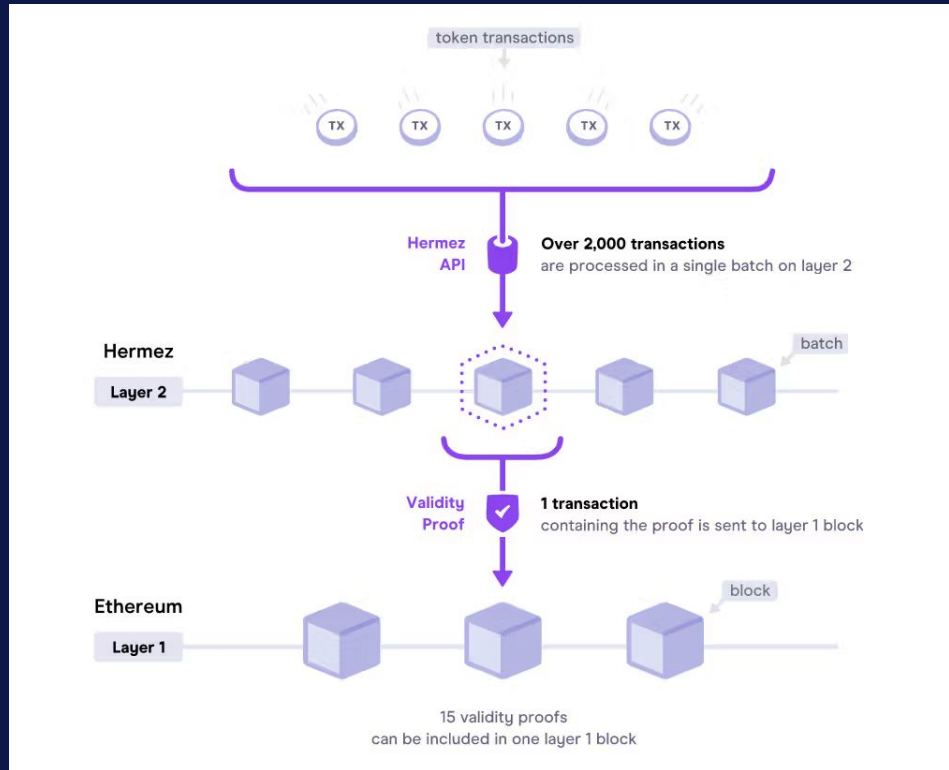
Lisk Blockchain

Lisk Blockchain



- Founded in 2016 by Max Kordek, and Oliver Beddows
- Part of [Ethereum ecosystem](#)
- A Growing [EVM Layer 2](#) (L2) Chain
- Part of [Optimism Superchain](#)
- [Near zero](#) fees

What is Layer 2?

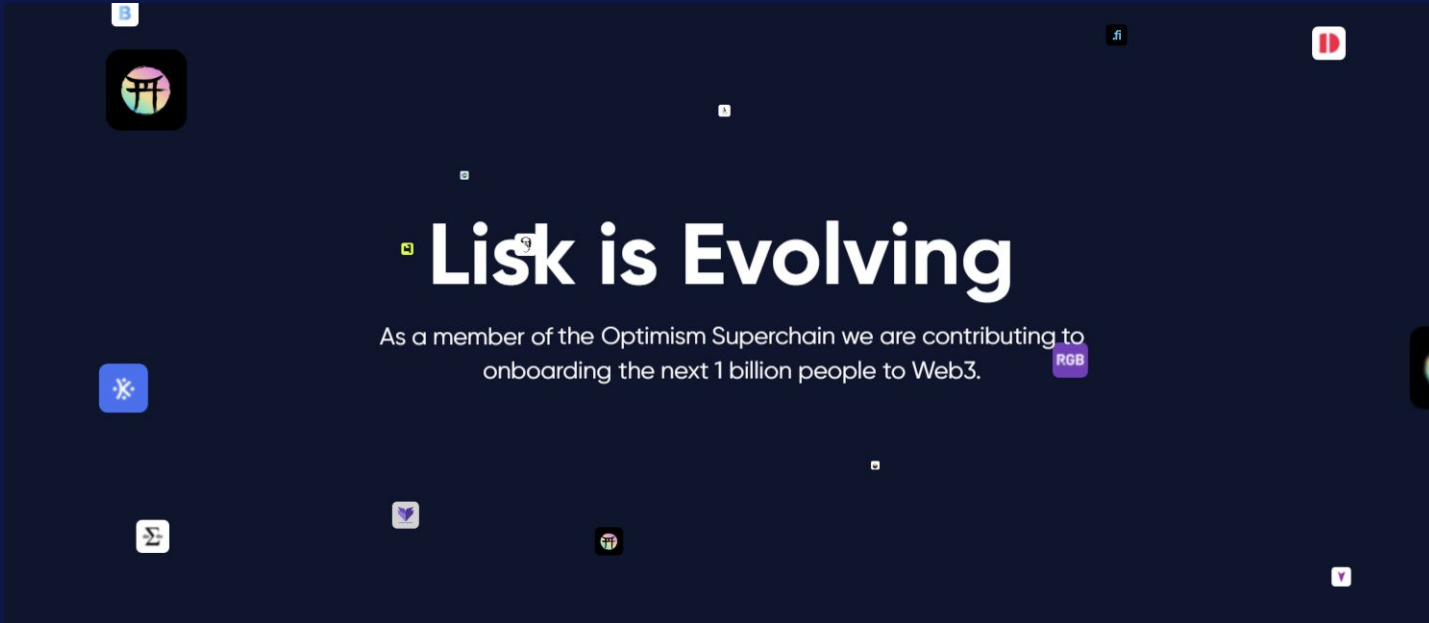


Sumber: Messari

Sumber: Thirdweb

Lisk Blockchain Website

Lisk is an L2 network focusing on real world solutions in emerging markets.



<https://lisk.com>

Lisk Blockchain Documentation

 Docs

Membangun dengan Lisk

Menggunakan Lisk

Indonesia

Q

Can

S

S

Selamat datang di dokumentasi Lisk 🎉

Jelajahi basis pengetahuan komprehensif kami yang dirancang khusus untuk pengembang dan pengguna! Temukan informasi teknis mendalam dan panduan bermanfaat tentang blockchain Lisk.

BUILT ON
the Superchain



Apa itu Lisk?

Lisk menawarkan jaringan Layer 2 (L2) yang sangat efisien, sangat cepat, dan mudah diskalakan yang dibangun di atas Optimism (OP) dan diamankan oleh Ethereum.

Jelajahi



Membangun dengan Lisk

Lisk menawarkan transaksi berbiaya rendah, kompatibilitas EVM, dan dukungan untuk semua framework pengembang utama, menjadikannya pilihan utama bagi para pengembang. Bergabunglah dengan Lisk untuk mengakses materi teknis penting bagi perjalanan pengembangan Anda.

Mulai



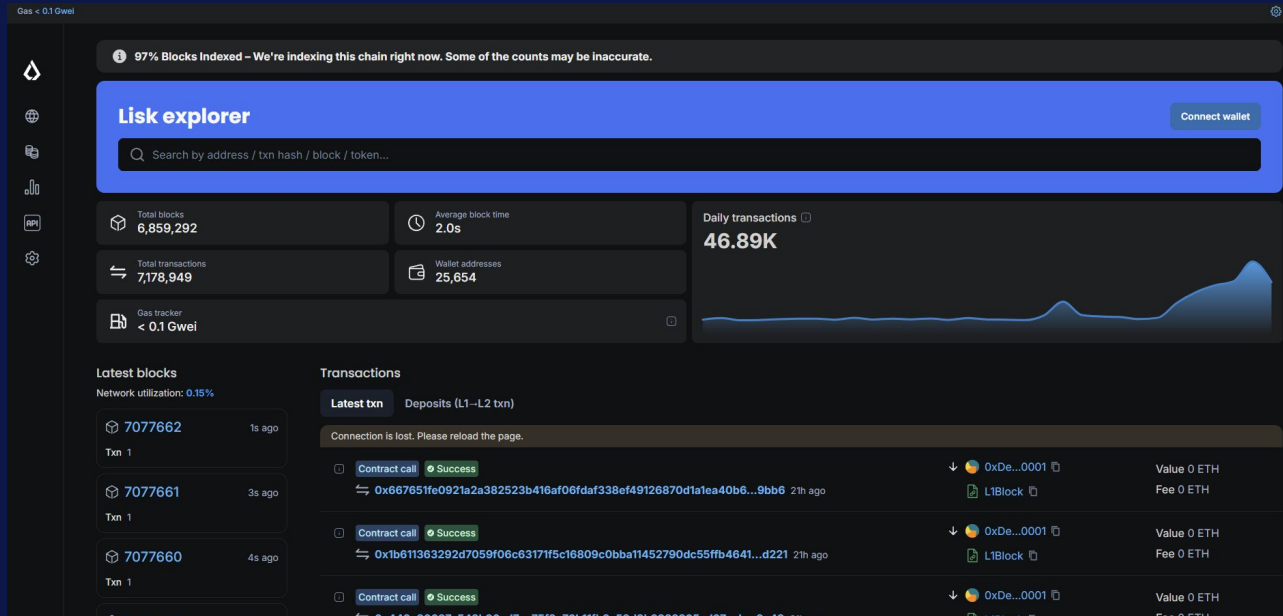
Ekosistem untuk masa depan

Lisk menggunakan framework superchain, menawarkan tata kelola yang terdesentralisasi dan ekosistem yang dapat dioperasikan bersama. Kami menyediakan peluang untuk mendukung pengembangan produk dan mengadvokasi prinsip-prinsip dunia yang terdesentralisasi dan inklusif.

Pelajari lebih lanjut

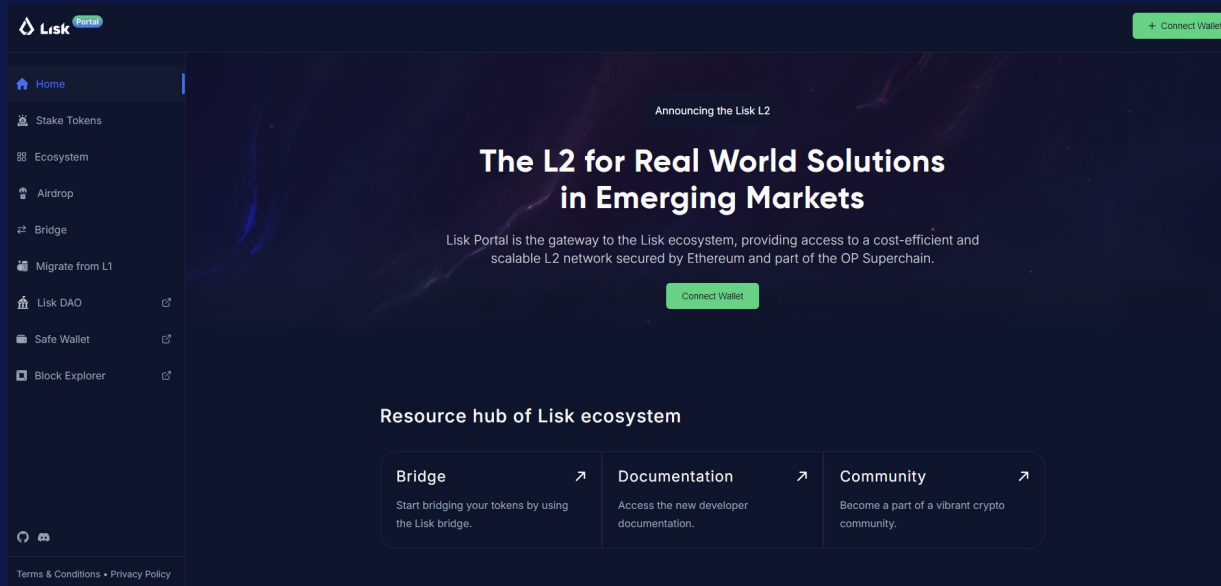
<https://docs.lisk.com/ind/>

Lisk Blockscout Blockchain Explorer



<https://blockscout.lisk.com/>

Lisk Portal



<https://portal.lisk.com/>

05

IDRX Stablecoin

What is Stablecoin?



A stablecoin is a type of cryptocurrency that is **pegged to a stable asset** (like the US dollar, gold, or a basket of assets) to reduce price volatility

What is IDRX?

IDRX is a blockchain-based digital currency, backed **1:1** by the **Indonesian Rupiah**, designed to offer **stability, transparency, and seamless** interaction with Web3 financial platforms.

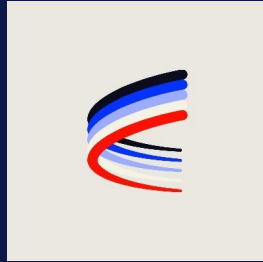


Supported Chain

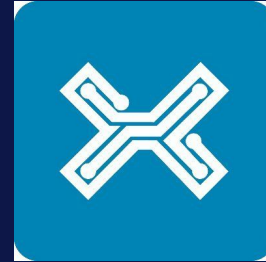


Enables **seamless interaction** with decentralized finance
(DeFi) platforms using IDRX

Where to get IDRX?



Aerodrome



Indodax

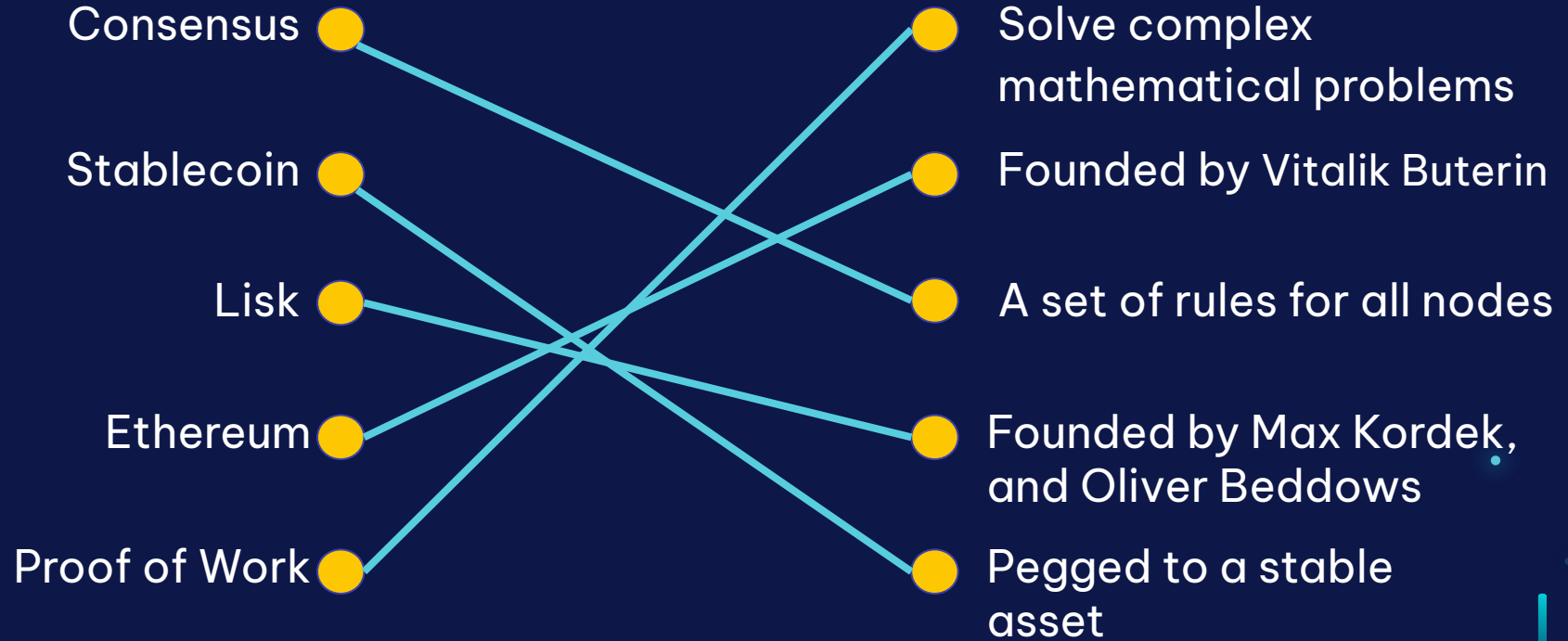


Uniswap



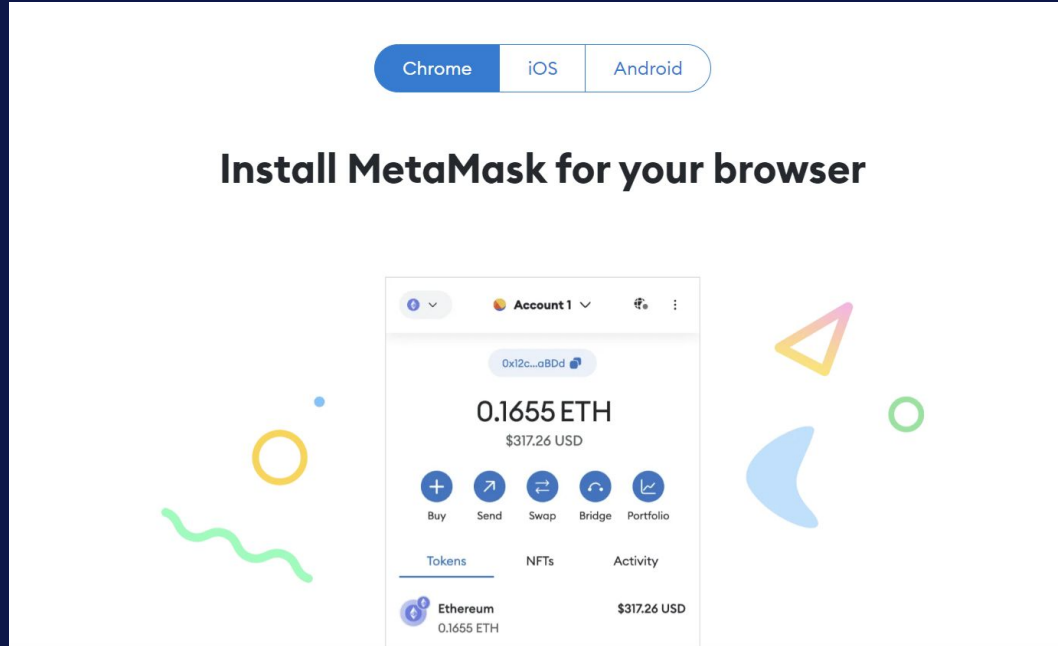
PancakeSwap

Quiz Time



Hands On

Install Metamask



Link: metamask.io/download

Create a Wallet



New to MetaMask?



No, I already have a Secret Recovery Phrase

Import your existing wallet using a Secret Recovery Phrase

Import wallet



Yes, let's get set up!

This will create a new wallet and Secret Recovery Phrase

Create a Wallet

Link: metamask.io/download

Add Lisk Sepolia Network

Announcing the Lisk L2

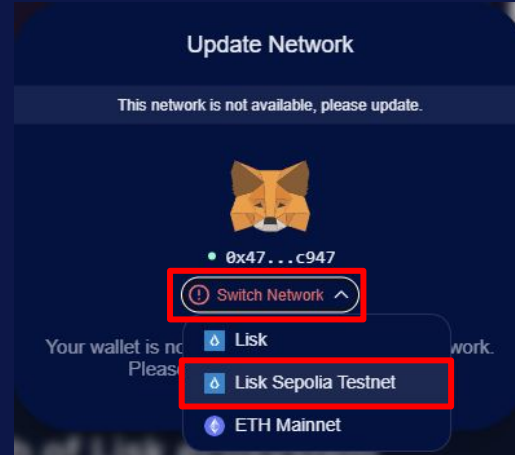
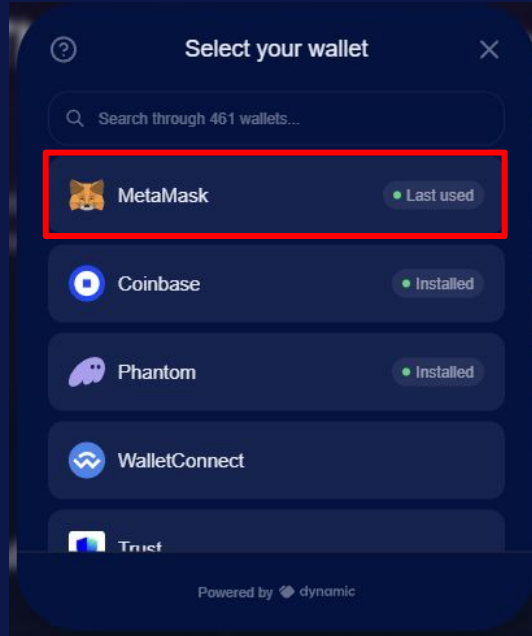
The L2 for Real World Solutions in Emerging Markets

Lisk Portal is the gateway to the Lisk ecosystem, providing access to a cost-efficient and scalable L2 network secured by Ethereum and part of the OP Superchain.

Connect Wallet

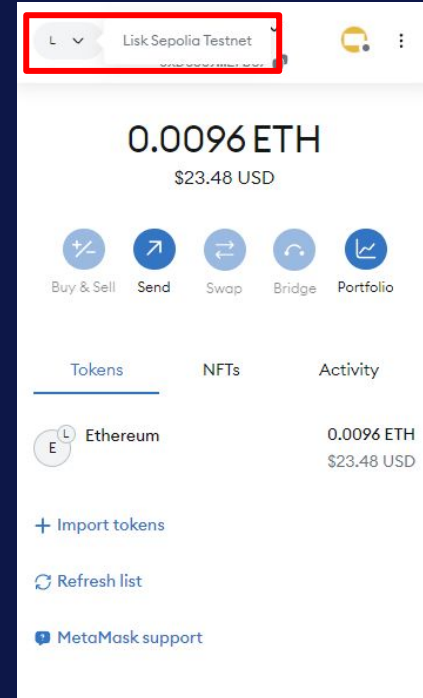
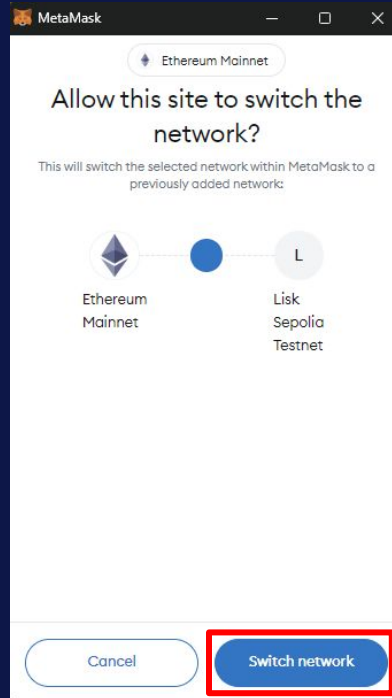
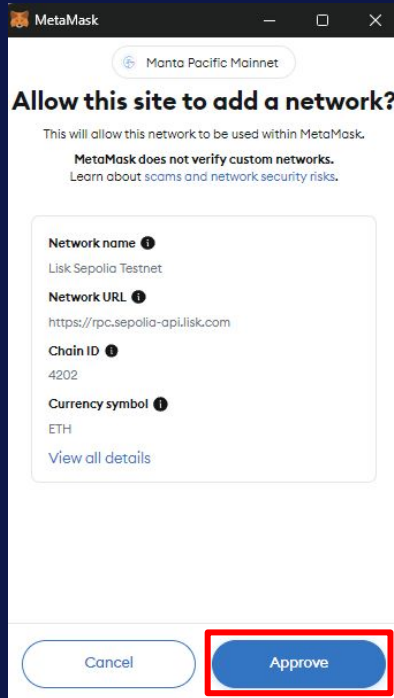
Link: <https://portal.lisk.com/>

Add Lisk Sepolia Network



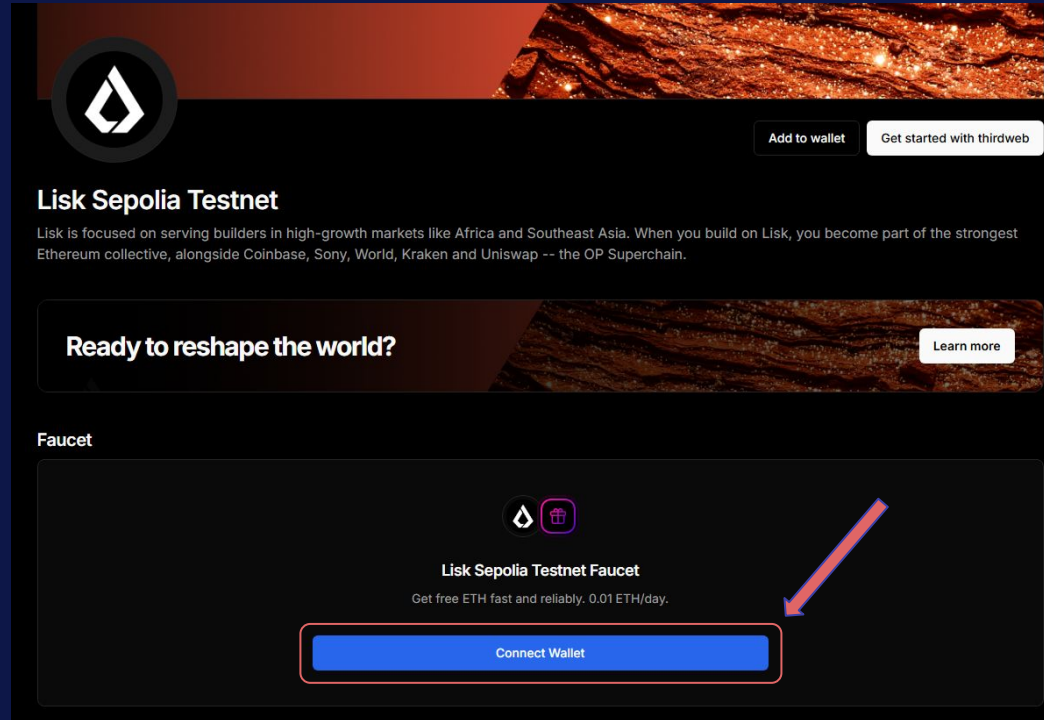
Link: <https://portal.lisk.com/>

Add Lisk Sepolia Network



Link: <https://portal.lisk.com/>

ETH Sepolia Faucet



The screenshot shows the Lisk Sepolia Testnet Faucet interface. At the top, there is a header with the Lisk logo and two buttons: "Add to wallet" and "Get started with thirdweb". Below the header, the text "Lisk Sepolia Testnet" is displayed, followed by a paragraph describing Lisk's focus on serving builders in high-growth markets. A section titled "Ready to reshape the world?" includes a "Learn more" button. The main section is titled "Faucet" and features a "Lisk Sepolia Testnet Faucet" heading, a subtext "Get free ETH fast and reliably. 0.01 ETH/day.", and a prominent blue "Connect Wallet" button. A red arrow points to the "Connect Wallet" button.

Lisk Sepolia Testnet

Lisk is focused on serving builders in high-growth markets like Africa and Southeast Asia. When you build on Lisk, you become part of the strongest Ethereum collective, alongside Coinbase, Sony, World, Kraken and Uniswap -- the OP Superchain.

Ready to reshape the world?

[Learn more](#)

Faucet

Lisk Sepolia Testnet Faucet

Get free ETH fast and reliably. 0.01 ETH/day.

[Connect Wallet](#)

Link: <https://thirdweb.com/lisk-sepolia-testnet>

ETH Sepolia Faucet

[Add to wallet](#) [Get started with thirdweb](#)

Lisk Sepolia Testnet ☆

Lisk is focused on serving builders in high-growth markets like Africa and Southeast Asia. When you build on Lisk, you become part of the strongest Ethereum collective, alongside Coinbase, Sony, World, Kraken and Uniswap -- the OP Superchain.

Ready to reshape the world?

[Learn more](#)

Faucet



Lisk Sepolia Testnet Faucet

Get free ETH fast and reliably. 0.01 ETH/day.

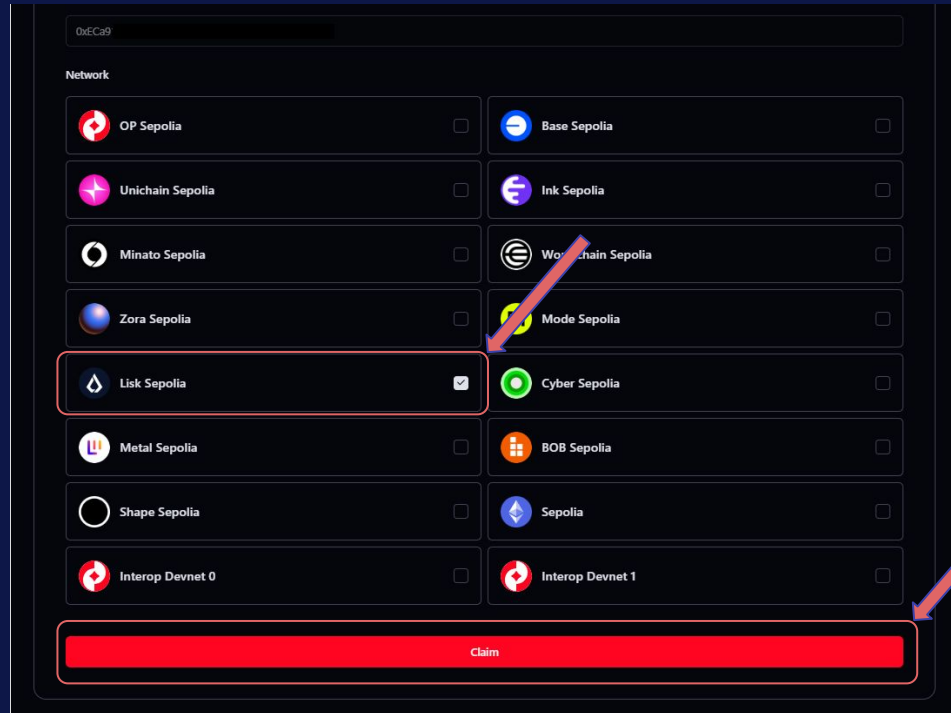
[Get 0.01 ETH](#)

☐ Verify you are human 

11,142 ETH left in the faucet

Link: <https://thirdweb.com/lisk-sepolia-testnet>

ETH Sepolia Faucet



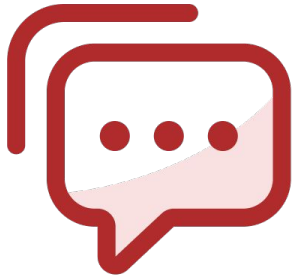
The screenshot shows the ETH Sepolia Faucet interface. At the top, there is a text input field containing the address '0xEc99'. Below this is a section titled 'Network' which contains a grid of 16 network selection options. Each option consists of a network icon, its name, and a checkbox. The 'Lisk Sepolia' option is selected, indicated by a checked checkbox and a red rectangular highlight. A red arrow points from the 'Lisk Sepolia' option to the 'Claim' button at the bottom of the interface. The 'Claim' button is a large red button with the word 'Claim' in white text. Another red arrow points from the bottom right of the interface towards the 'Claim' button.

Network	Selected
OP Sepolia	☐
Base Sepolia	☐
Unichain Sepolia	☐
Ink Sepolia	☐
Minato Sepolia	☐
Wormchain Sepolia	☐
Zora Sepolia	☐
Mode Sepolia	☐
Lisk Sepolia	☒
Cyber Sepolia	☐
Metal Sepolia	☐
BOB Sepolia	☐
Shape Sepolia	☐
Sepolia	☐
Interop Devnet 0	☐
Interop Devnet 1	☐

Claim

Link: <https://console.optimism.io/faucet>

Q & A



Audience Q&A

① Click **Present with Slido** or install our
while presenting.

to show live Q&A

Foto Bersama

Thank You



**Pelita Bangsa
Academy**

PT Pelita Teknologi Bangsa

Mall of Indonesia, Lantai 2, Unit 2F-A7B,
Jalan Boulevard Barat Raya No. 1,
Kelapa Gading Jakarta Utara 14240.



pelitabangsa.co.id



[instagram.pelitabangsa.co.id](https://www.instagram.com/pelitabangsa.co.id)



[linkedin.pelitabangsa.co.id](https://www.linkedin.com/company/pelitabangsa.co.id)



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